Practice workbook · Describe fertilisation and early embryo development, including twins

From egg release through gamete fusion to mitotic division of the zygote, including how identical twins arise.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SEQUENCE-FERTILISATION-AND-EARLY-EMBRYO

# 1. Question 6 · part b · 4 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b only.

**6** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

**(2)**

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# 1. Question 6 · part b · 4 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

**(4)**

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(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

**(1)**

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

## 1. Question 6 · part b · 4 marks · Working and answer

Working and answer

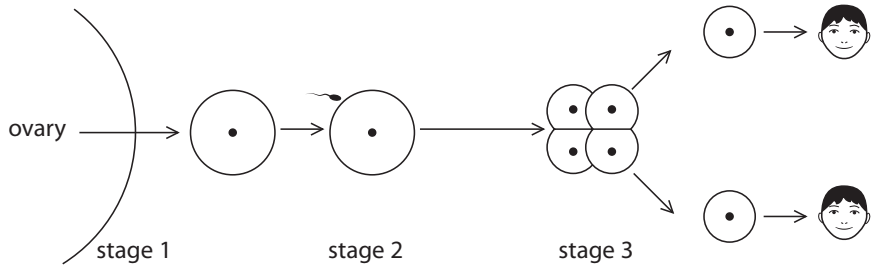
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## 2. Question 5 · part a.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.i only.

**5** (a) The diagram shows how genetically identical twins are produced.




(i) Name the process that occurs at stage 1.

**(1)**

(ii) Name the process that occurs at stage 2.

**(1)**

(iii) Use information from the diagram to explain how genetically identical twins are formed.

**(2)**

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## 2. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

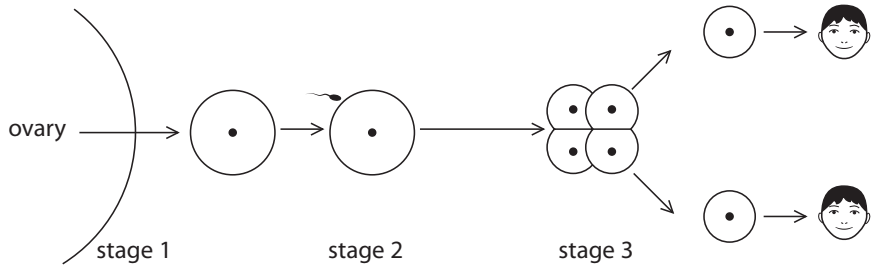
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# 3. Question 5 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

**5** (a) The diagram shows how genetically identical twins are produced.




(i) Name the process that occurs at stage 1.

**(1)**

(ii) Name the process that occurs at stage 2.

**(1)**

(iii) Use information from the diagram to explain how genetically identical twins are formed.

**(2)**

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## 3. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

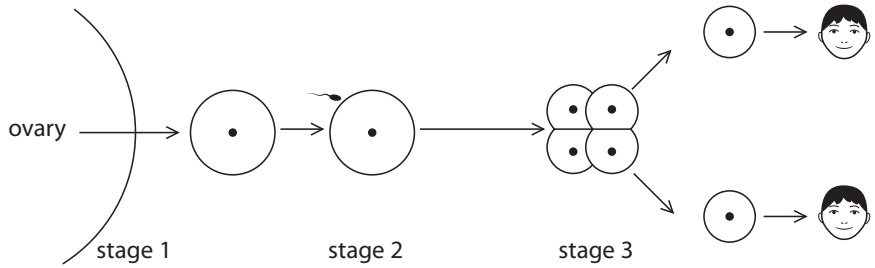
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# 4. Question 5 · part a.iii · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

**5** (a) The diagram shows how genetically identical twins are produced.




(i) Name the process that occurs at stage 1.

**(1)**

(ii) Name the process that occurs at stage 2.

**(1)**

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**(2)**

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# 4. Question 5 · part a.iii · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

(b) (i) In-vitro fertilisation (IVF) increases the chance of producing twins.

Which hormones are given to a female at the start of IVF treatment?

**(1)**

- ☒ **A** FSH and LH
- ☒ **B** LH and oestrogen
- ☒ **C** oestrogen and progesterone
- ☒ **D** progesterone and FSH

(ii) Describe the procedures in IVF after hormone treatment.

**(4)**

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**(Total for Question 5 = 9 marks)**

## 4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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## 5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

**8** (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

**(1)**

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

**(3)**

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**24**

## 5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

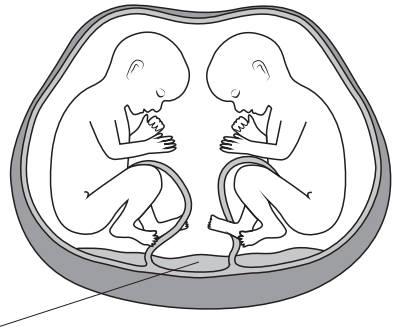
Further practice · Attempt part b.ii only.

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(b) Some identical twins share a placenta, as shown in the diagram.




(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

**(3)**

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(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

**(2)**

# 5. Question 8 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.




(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

**(1)**

percentile = .....

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## 5. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

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# 6. Question 6 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

**6** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

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# 6. Question 6 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

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**(4)**

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**(1)**

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- ☒ **C** the cells are able to differentiate into any type of sex cell
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## 6. Question 6 · part a · 2 marks · Working and answer

Working and answer

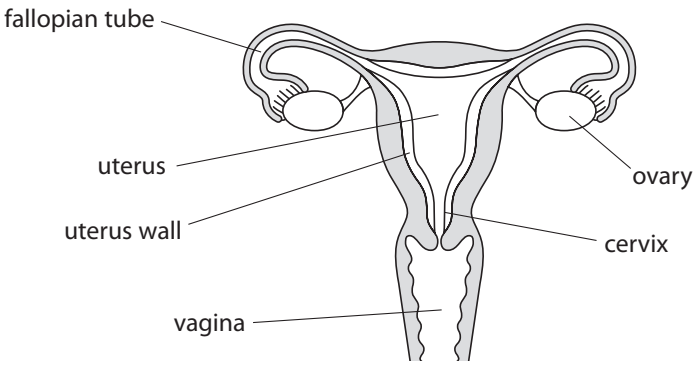
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# 7. Question 2 · part a.iii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

**2** (a) The diagram shows the female reproductive system.




(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

**(3)**

| Part | Role | | |
|---|---|---|---|
| | where sperm are deposited | where eggs are produced | where fertilisation takes place |
| fallopian tube | | | |
| ovary | | | |
| vagina | | | |

(ii) Describe the action of the muscles in the uterus wall during labour.

**(2)**

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# 7. Question 2 · part a.iii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

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(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

**(1)**

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

**(1)**

.....

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## 7. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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